

27.4.20

Tutorial -3

A1. Let $\overline{v_{LNA}}$ be the i/f referred noise of the LNA.

$$\therefore F_1 = 1 + \frac{A_1^2 \times \overline{v_{LNA}}}{A_1^2 \times \overline{v_{AS}}}$$

$$\therefore \overline{v_{LNA}} = \overline{v_{AS}} (F_1 - 1)$$

Let A_2 be the gain of the mixer.

$$\therefore \overline{v_{no}} = A_1 A_2 (\overline{v_{AS}} + \overline{v_{LNA}}) + A_2 (\overline{v_{nI}} + \overline{v_{nQ}})$$

$$\overline{v_{nI, AS}} = A_1 A_2 \overline{v_{AS}}$$

$$\text{where } \overline{v_{nI}} = \overline{v_{nQ}} = \overline{v_n^2}$$

$$\therefore F_{net} = \frac{A_1^2 A_2^2 \overline{v_{AS}} + A_1^2 A_2^2 \overline{v_{LNA}} + A_2^2 (\overline{v_n^2} + \overline{v_n^2})}{A_1^2 A_2^2 \overline{v_{AS}}}$$

$$= 1 + \frac{\overline{v_{LNA}}}{\overline{v_{AS}}} + \frac{2 \overline{v_n^2}}{A_1^2 \overline{v_{AS}}}$$

$$F_{net} = F_1 + \frac{2 \overline{v_n^2}}{A_1^2 \times 4 \times 7 \times 10^9}$$

4.2. $H_1(s) = H_2(s) = -\gamma_m \times 2$

where $\frac{1}{2} = \frac{1}{\alpha_x} + sC_1 + \frac{1}{sL_1}$

$$\therefore H_1(s) = H_2(s) = \frac{-\gamma_m \times sL_1 \alpha_x}{s^2 L_1 C_1 \alpha_x + sL_1 + \alpha_x}$$

Each contributes $180^\circ \Rightarrow$ denominator contributes $+90^\circ$

$$H(s) = H_1(s) \times H_2(s) = \frac{\gamma_m \gamma_m s^2 L_1 L_2 \alpha_x^2}{(s^2 L_1 C_1 \alpha_x + sL_1 + \alpha_x)(s^2 L_2 C_1 \alpha_x + sL_2 + \alpha_x)}$$

$\therefore (s^2 L_1 C_1 \alpha_x + sL_1 + \alpha_x)(s^2 L_2 C_1 \alpha_x + sL_2 + \alpha_x)$
contributes 180°

$$\therefore \tan^{-1} \left(\frac{\omega L_1}{\alpha_x - \omega^2 L_1 C_1 \alpha_x} \right) + \tan^{-1} \left(\frac{\omega L_2}{\alpha_x - \omega^2 L_2 C_1 \alpha_x} \right) = 180^\circ$$

$$\therefore \frac{\omega L_1}{\alpha_x - \omega^2 L_1 C_1 \alpha_x} + \frac{\omega L_2}{\alpha_x - \omega^2 L_2 C_1 \alpha_x} = 0$$

$$L_1 \alpha_x - \omega^2 L_1 L_2 C_1 \alpha_x + L_2 \alpha_x - \omega^2 L_1 L_2 C_1 \alpha_x = 0$$

$$\therefore \omega^2 = \frac{L_1 + L_2}{2C_1 (L_1 L_2)}$$

4. If $L_2 = L_1 + 0L \therefore \omega^2 = \frac{L_1 + 0L}{C_1 L_1 (L_1 + 0L)}$

$$\omega^2 = \frac{1}{L_1 C_1} \left[\frac{1 + \frac{\sigma L}{2L_1}}{1 + \frac{\sigma L}{L_1}} \right] \approx \frac{1}{L_1 C_1} \left[1 + \frac{\sigma L}{2L_1} \right] \left[1 - \frac{\sigma L}{L_1} \right]$$

$$\therefore \omega^2 \approx \frac{1}{L_1 C_1} \left[1 - \frac{\sigma L}{2L_1} \right] \Rightarrow \omega \approx \frac{1}{\sqrt{L_1 C_1}} \left[1 - \frac{\sigma L}{4L_1} \right]$$

Q If $L_1 = L + \frac{\sigma L}{2}$, $L_2 = L - \frac{\sigma L}{2}$

$$\omega^2 = \frac{L}{C_1 (L^2 - \frac{\sigma L^2}{4})} = \frac{1}{LC} \times \frac{1}{1 - \frac{\sigma L^2}{4L^2}} \approx \frac{1}{LC} \left[1 + \frac{\sigma L^2}{4L^2} \right]$$

$$\omega \approx \frac{1}{\sqrt{LC}} \left[1 + \frac{\sigma L^2}{8L^2} \right]$$

A-3. $\phi = -180^\circ + 90^\circ - \tan^{-1} \left[\frac{\omega L}{\alpha(1 - \omega^2 LC)} \right]$

$$\therefore \frac{d\phi}{d\omega} = \frac{1}{1 + \left[\frac{\omega L}{\alpha(1 - \omega^2 LC)} \right]^2} \times \frac{\alpha(1 - \omega^2 LC)L + \omega L^2 \omega \alpha C}{[\alpha(1 - \omega^2 LC)]^2}$$

$$\frac{d\phi}{d\omega} = \frac{L\alpha(1 - \omega^2 LC) + 2\omega^2 \alpha L^2 C}{[\alpha(1 - \omega^2 LC)]^2 + \omega^2 L^2}$$

$$\frac{d\phi}{d\omega} \bigg|_{\omega_0} = \frac{2\omega_0^2 \alpha L^2 C}{\omega_0^2 L^2} = \frac{2\alpha C}{1} = 2\alpha C$$

→ $\phi = \frac{\omega_0}{2} \frac{d\phi}{d\omega} = \frac{\omega_0}{2} \times 2\alpha C = \omega_0 \alpha C = \frac{\alpha C}{\sqrt{LC}} = \frac{\alpha \sqrt{LC}}{L}$

$$\therefore \phi = \frac{\alpha \sqrt{LC}}{L}$$